

Cognizant GenC Next – Complete 40-Week Placement Roadmap

For a 3-1 CSE (Cybersecurity) student | GenC Next + Cybersecurity + Cloud Security

Important: Cognizant's exact campus assessment can vary by role and drive. The current MITS placement notification should be treated as the final authority for the specific drive.

How to use this roadmap

Use the first 40 weeks as the main preparation plan. Maintain a weekly mistake notebook, solve problems under time limits, build projects as you learn, and revise weak topics every Sunday.

Phase 1 — Coding Foundation — Weeks 1–4

Week	Skills / Deliverables	Free Resources
1	Python fundamentals: variables, data types, I/O, operators, conditions, loops	Python.org Tutorial; CS50P; freeCodeCamp
2	Strings, lists, tuples, sets, dictionaries	Python.org; freeCodeCamp; Programiz
3	Functions, parameters, return values, scope, recursion basics	CS50P; Python.org
4	OOP: classes, objects, constructors, inheritance, polymorphism, encapsulation, abstraction	freeCodeCamp; Programiz

Phase 2 — DSA — Weeks 5–12

Week	Skills / Deliverables	Free Resources
5	Arrays, traversal, insertion/deletion, two pointers	GeeksforGeeks; LeetCode
6	Strings, palindrome, anagram, frequency, substrings	GeeksforGeeks; LeetCode
7	Linear and binary search	GeeksforGeeks; VisuAlgo
8	Bubble, selection, insertion, merge-sort concept	GeeksforGeeks; VisuAlgo
9	Stack and queue	GeeksforGeeks; HackerRank
10	Linked lists	GeeksforGeeks; HackerRank
11	Hashing: maps/sets and frequency problems	GeeksforGeeks; LeetCode
12	Recursion + mixed DSA mock	LeetCode; HackerRank

Phase 3 — SQL + DBMS — Weeks 13–16

Week	Skills / Deliverables	Free Resources
13	SELECT, WHERE, ORDER BY, DISTINCT, aggregate functions	SQLBolt; SQLZoo
14	GROUP BY, HAVING, subqueries	SQLBolt; SQLZoo
15	INNER/LEFT/RIGHT/SELF JOIN	SQLBolt; HackerRank SQL
16	Keys, normalization, transactions, ACID, indexing	Neso Academy; Gate Smashers

Phase 4 — Core CS — Weeks 17–22

Week	Skills / Deliverables	Free Resources
17	OOP revision	freeCodeCamp; Programiz
18	OS: processes, threads, scheduling, deadlocks, memory	Neso Academy; Gate Smashers
19	Networks: OSI, TCP/IP, TCP vs UDP, DNS, DHCP, HTTP/HTTPS	Cisco Networking Academy; Practical Networking
20	IP addressing, subnetting, routing, NAT, ports, firewalls	Practical Networking; Cisco Networking Academy
21	DBMS revision + SQL mixed practice	Neso Academy; SQLBolt; HackerRank
22	Mixed core-CS technical test + weak-area revision	GeeksforGeeks interview prep; your notes

Phase 5 — Cybersecurity + Cloud — Weeks 23–28

Week	Skills / Deliverables	Free Resources
23	CIA triad, threats, vulnerabilities, risk, authentication, authorization	Cisco Introduction to Cybersecurity; TryHackMe free content
24	Encryption, hashing, digital signatures, certificates	Crypto 101; OWASP materials
25	Linux: users, groups, permissions, processes, SSH, logs	Linux Journey; OverTheWire Bandit
26	Web security: SQLi, XSS, CSRF, access control, OWASP concepts	OWASP WSTG; PortSwigger Web Security Academy
27	Cloud: IaaS/PaaS/SaaS, deployment models, shared responsibility	AWS Skill Builder; Microsoft Learn
28	IAM, VPC, security groups, encryption, logging/monitoring	AWS Skill Builder; AWS docs; Microsoft Learn

Phase 6 — Projects + Resume — Weeks 29–32

Week	Skills / Deliverables	Free Resources
29	Project 1: Password Strength Checker	Python.org; GitHub Skills
30	Project 2: Cloud Security Monitoring System	AWS Skill Builder; AWS docs; GitHub Skills
31	README, architecture diagram, screenshots, setup instructions, GitHub	GitHub Skills; Markdown Guide
32	Project interview preparation and mock explanation	Project README; record and review yourself

Phase 7 — GenC Next Intensive — Weeks 33–36

Week	Skills / Deliverables	Free Resources
33	20 coding + 20 SQL + 50 aptitude questions	LeetCode; HackerRank; IndiaBix
34	Arrays, strings, hashing, searching/sorting + SQL joins	LeetCode; HackerRank SQL
35	OOP, OS, DBMS, networks mixed revision	Neso Academy; Gate Smashers; Cisco
36	Full mixed mock: aptitude + coding + SQL + technical	HackerRank; LeetCode; IndiaBix

Phase 8 — Interview — Weeks 37–40

Week	Skills / Deliverables	Free Resources
37	HR: self-introduction, why Cognizant, why cybersecurity, strengths/weaknesses, goals	Cognizant Careers interview resources; STAR practice
38	Project interview: explain both projects without memorizing	Your project README + mock recordings
39	Technical interview: Python/DSA/SQL/OOP/OS/DBMS/networks/cyber/cloud	Revision notes + GFG/HackerRank
40	Full placement simulation: aptitude → coding → technical → project → HR	All above practice platforms

Final GenC Next + Cybersecurity Placement Readiness

The 40-week roadmap is a strong preparation framework, but for a cybersecurity placement you should reach the following readiness level before the drive:

Area	Target before the drive
Python	Strong fundamentals + interview coding
DSA	100–150 quality problems; focus on arrays, strings, hashing, search, sorting, stack/queue, linked lists, recursion
SQL	100+ queries; joins, grouping, subqueries and DBMS concepts
OOP	Interview-ready
DBMS	Keys, normalization, ACID, transactions, indexing
OS	Processes, threads, scheduling, deadlocks, memory
Networks	OSI/TCP-IP, TCP/UDP, DNS, HTTP/HTTPS, IP, subnetting, routing, ports, firewalls
Linux	Practical command-line use, permissions, processes, SSH and logs
Cybersecurity	Strong fundamentals + practical labs; OWASP/web-security basics
Cloud	AWS/Azure fundamentals + IAM, VPC, security groups, encryption and logging
Projects	2 strong, working and documented cybersecurity/cloud projects
Git/GitHub	Projects documented with README, screenshots and useful commit history
Resume	One-page, cybersecurity-focused fresher resume
Aptitude	Regular timed practice
Communication	1-minute introduction + clear technical explanations
Interview	Repeated technical, project and HR mock interviews

Recommended Cybersecurity Resume

Header: Lakshmi Sai Reddy — Cybersecurity | Cloud Security

Contact: Phone | Professional email | LinkedIn | GitHub

Education: B.Tech – CSE (Cybersecurity), MITS Deemed to be University | CGPA: 8.2

Technical Skills: Python, C, SQL, Linux, Networking, DBMS, OS, OOP, Cybersecurity Fundamentals, OWASP/Web Security, Cryptography Fundamentals, AWS/Azure, IAM, VPC, Security Groups, Git/GitHub.

Projects: 1) Password Strength Checker 2) Cloud Security Monitoring System 3) Optional: Network Vulnerability Scanner.

Certifications: Include only relevant certifications that you actually completed.

Important resume rule: Do not list a technology you cannot explain in an interview. Keep the fresher resume to one page unless the placement instructions say otherwise.

Suggested Preparation Routine

Monday–Friday (4 hours/day): 1.5 h coding/DSA + 45 min SQL + 45 min aptitude + 1 h core CS/cyber/cloud.

Saturday: revision + coding/SQL test + communication practice.

Sunday (6 hours): 2 h DSA + 1 h SQL + 1 h aptitude + 1 h cybersecurity/cloud + 1 h mock test and mistake review.

Final Drive Checklist

■ 100–150 coding problems solved ■ 100+ SQL queries ■ 2 working cybersecurity/cloud projects ■ GitHub updated ■ One-page resume ready ■ Aptitude mocks practiced ■ Self-introduction ready ■ Project explanation ready ■ Core-CS technical questions revised ■ Cybersecurity questions revised ■ At least 3 full placement simulations completed

Remember: this roadmap prepares you broadly for GenC Next and cybersecurity-oriented placements. The actual Cognizant/MITS drive notification remains the final source for eligibility, assessment pattern, dates and role-specific requirements.